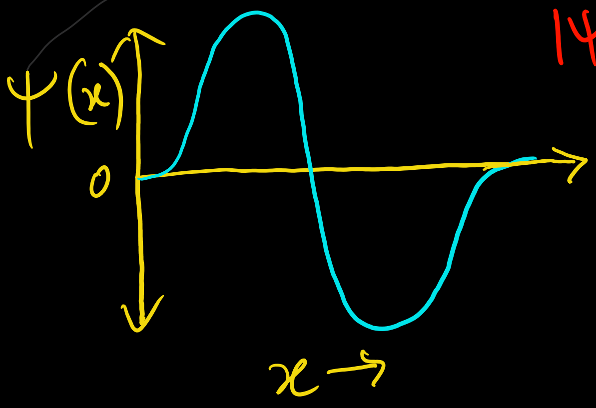
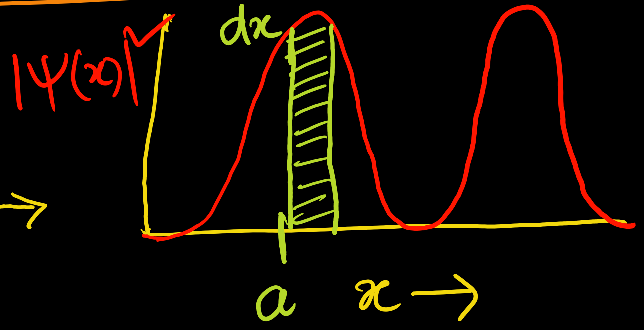
Interpreting the wavefunction





$\Psi(x) \longrightarrow$ wave function

magnitude

$|\Psi(x)|^2 = \Psi^*(x)\Psi(x) \longrightarrow$ Probability Density

Complex conjugate of Si

$\Psi(x) \in \mathbb{C}$ in general Complex number

$\Psi(x) = Re(x) + i\, Im(x)$

$\Psi^*(x) = Re(x) - i\, Im(x)$

$\Psi^*(x)\Psi(x) = Re^2(x) + Im^2(x)$

$c = a + bi$

real imaginary

$i = \sqrt{-1}$

$c^* = a - bi$

$|c^2| = c^* c$

"Born Interpretation"

$\Psi^*(x)\Psi(x)\,dx =$ Probability that the particle is located between x and $x + dx$

$$\boxed{\int_{-\infty}^{\infty} \Psi^*(x)\Psi(x)\,dx = 1}$$

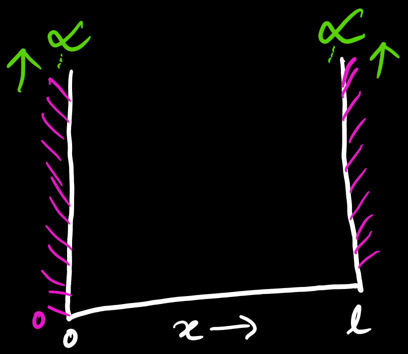
"Normalization"

Wave function must be normalized

Particle in a Box

Free Particle,

$$V(x) = \begin{cases} 0 & \text{if } 0 < x < l \\ \infty & \text{otherwise} \end{cases}$$



V = 0 in box and wall
V = ∞ outside

$$\hat{H}\psi(x) = E\psi(x)$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \cancel{V(x)}\psi(x) = E\psi(x)$$

0

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2m}{\hbar^2}E\psi(x)$$

$$\psi(x) = A\cos(kx) + B\sin(kx)$$

where $k = \frac{\sqrt{2mE}}{\hbar}$

$$\psi(x \leq 0) = 0 \longrightarrow \psi(0) = 0$$

$$\psi(x \geq l) = 0 \longrightarrow \psi(l) = 0$$

$$\psi(0) = A\cos(k.0) + B\sin(k.0) = A \rightarrow A = 0$$

$$\psi(l) = B\sin(kl) = 0 \qquad kl = \sin^{-1}(0)$$

$$= n\pi$$

$$k = \frac{n\pi}{l} \qquad n \in \mathbb{Z}$$

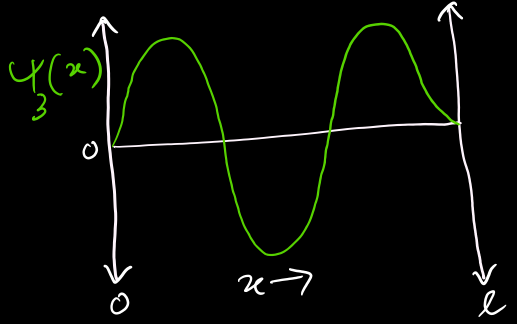
$$\boxed{\psi_n(x) = B\sin\left(\frac{n\pi x}{l}\right)} \qquad n = 1, 2, 3 \ldots$$

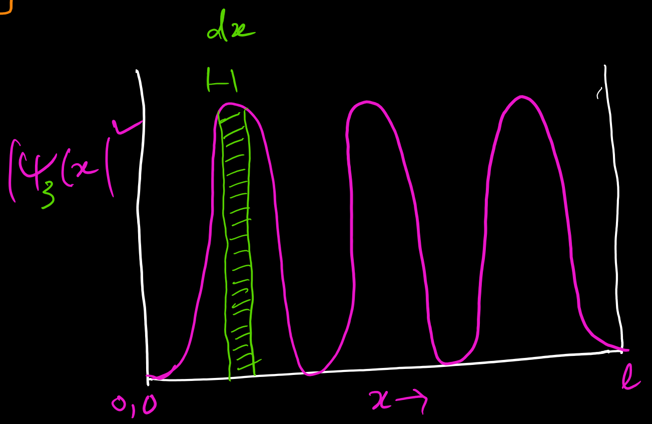
$$\frac{n\pi}{l} = \frac{\sqrt{2mE}}{\hbar} \qquad \frac{n^2\hbar^2\pi^2}{l^2} = 2mE$$

$$\boxed{E_n = \frac{h^2 n^2}{8ml^2}} \qquad n = 1, 2, 3 \ldots$$

$n \rightarrow$ Quantum Number

Normalization





$$\psi_n(x) = B \sin\left(\frac{n\pi x}{\ell}\right) \qquad E_n = \frac{h^2 n^2}{8m\ell^2} \quad n = 1, 2 \ldots$$

How to get B?

$$|\psi_n(x)|^2 dx = \psi_n^*(x)\psi_n(x)\,dx = \text{Probability}$$

$$x \leq (\text{Particle position}) \leq x + dx$$

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)\,dx = 1 \qquad \text{——— (1)}$$

$$\psi(x) = 0 \quad \text{if } x \leq 0 \text{ or } x \geq \ell$$

$$\Rightarrow 1 = \int_0^\ell [B\sin(n\pi x/\ell)]^*[B\sin(n\pi x/\ell)]\,dx$$

$$\Rightarrow 1 = B^2\int_0^\ell \sin^2(n\pi x/\ell)\,dx$$

$$\Rightarrow \boxed{\int \sin^2(kx) = \frac{1}{2}(1 - \cos(2kx))}$$ we know this

$$\text{So, } 1 = \frac{B^2}{2}\int_0^\ell (1 - \cos(2n\pi x/\ell))\,dx$$

$$\text{or, } \frac{2}{B^2} = \left(x - \frac{\ell}{2\pi n}\sin\left(\frac{2\pi n x}{\ell}\right)\right)\Big|_0^\ell$$

$$\text{or, } \frac{2}{B^2} = \left[\ell - \frac{\ell}{2\pi n}\cancel{\sin(2\pi n)}\right] - \left[0 - \cancel{\frac{\ell}{2\pi}\sin(0)}\right]$$

$$\text{or, } \frac{2}{B^2} = \ell \quad \text{or } B = \sqrt{\frac{2}{\ell}}$$

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{\ell}}\sin\left(\frac{n\pi x}{\ell}\right)}$$

$B \to$ Normalization constant

Wave function plot

$$\psi_n = \sqrt{\frac{2}{l}} \sin\left(\frac{n\pi x}{l}\right)$$

At Ground state, $n = 1$

$$\psi_1 = \sqrt{\frac{2}{l}} \sin\left(\frac{\pi x}{l}\right)$$

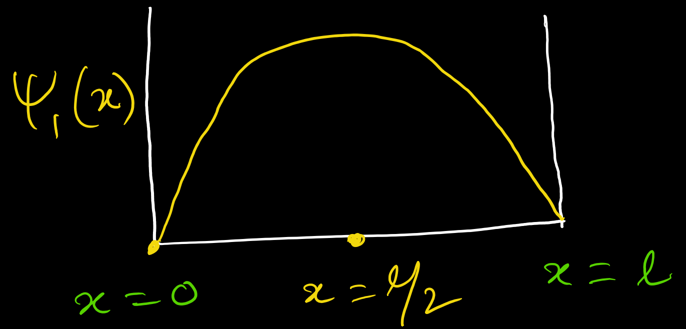
$\sin 0^\circ \longrightarrow \sin 90^\circ$

$0 \quad \frac{1}{2} \quad \frac{1}{\sqrt{2}} \quad \frac{\sqrt{3}}{2} \quad 1$

For max. value of ψ_1

$$\sin\left(\frac{\pi x}{l}\right) = 1 = \sin\frac{\pi}{2}$$

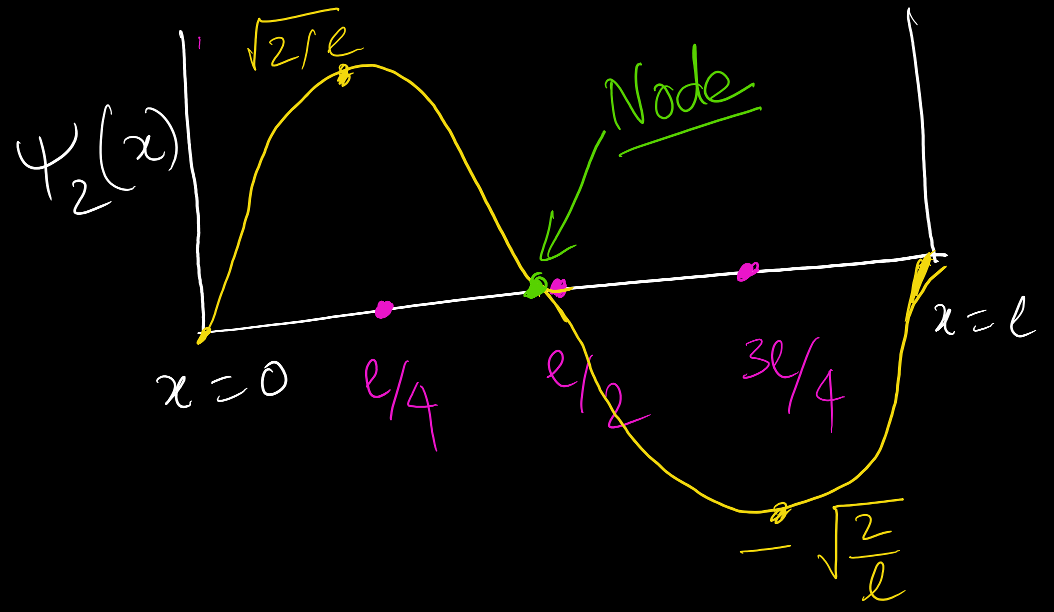
So, $\frac{\pi x}{l} = \frac{\pi}{2}$; or, $\boxed{x = \frac{l}{2}}$



First excited State, $n = 2$ $\quad \psi_2 = \sqrt{\frac{2}{l}} \sin\left(\frac{2\pi x}{l}\right)$, At what x, ψ_2 is max?

$$\sin\left(\frac{2\pi x}{l}\right) = 1 = \sin\frac{\pi}{2}$$

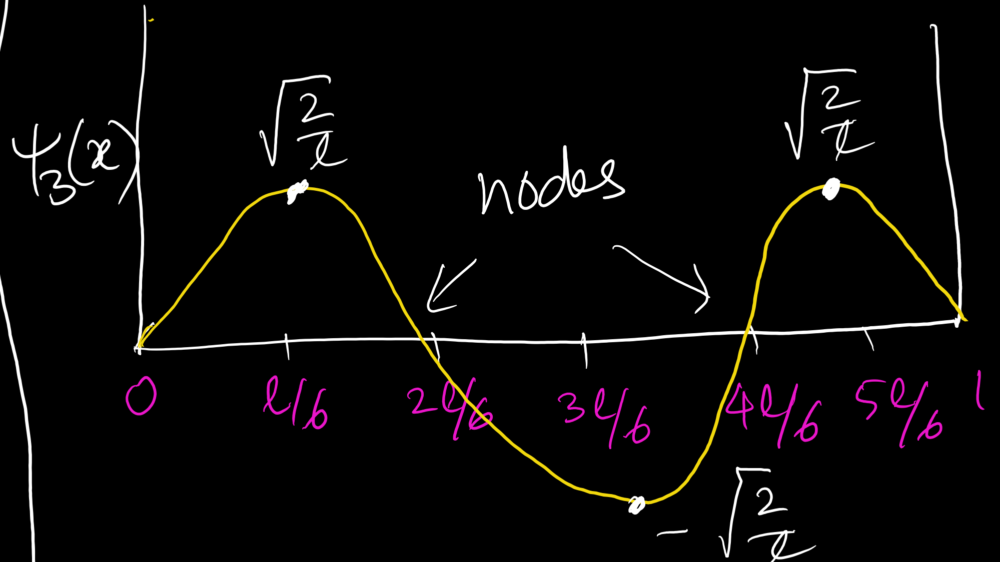
$\Rightarrow \frac{2x}{l} = \frac{1}{2}$ or, $\boxed{x = l/4}$



Second excited State : $n = 3$

$\boxed{x = l/6}$

ES2 —— $n = 3$
ES1 —— $n = 2$
GS —— $n = 1$



Wave Function Plot 2

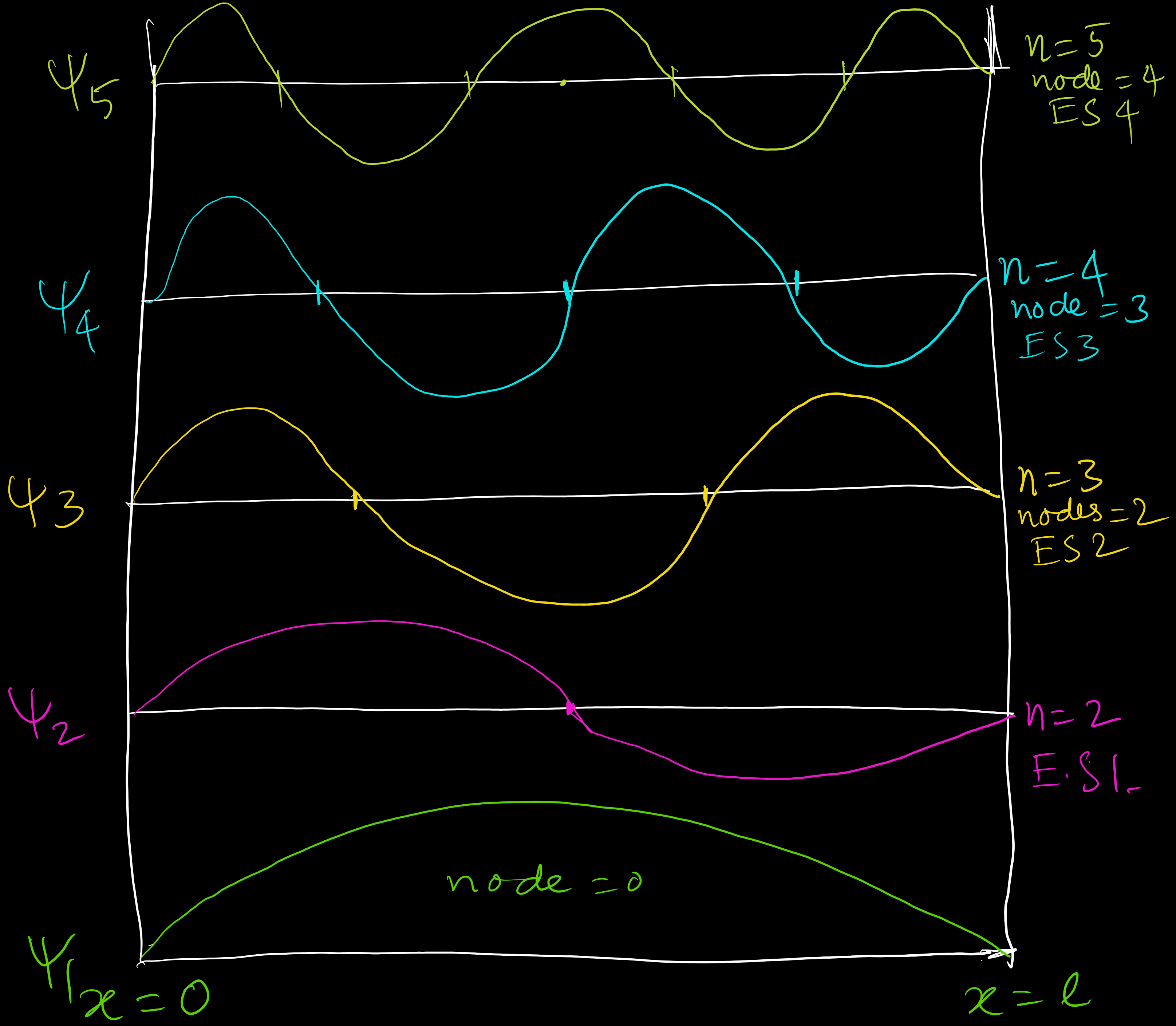
Plot of ψ_{10} ?

- at $\left.\begin{matrix} x = 0 \\ x = l \end{matrix}\right\} \Rightarrow \psi = 0$
- Nodes $= (n-1)$

So for ψ_{10}, $n = 10$

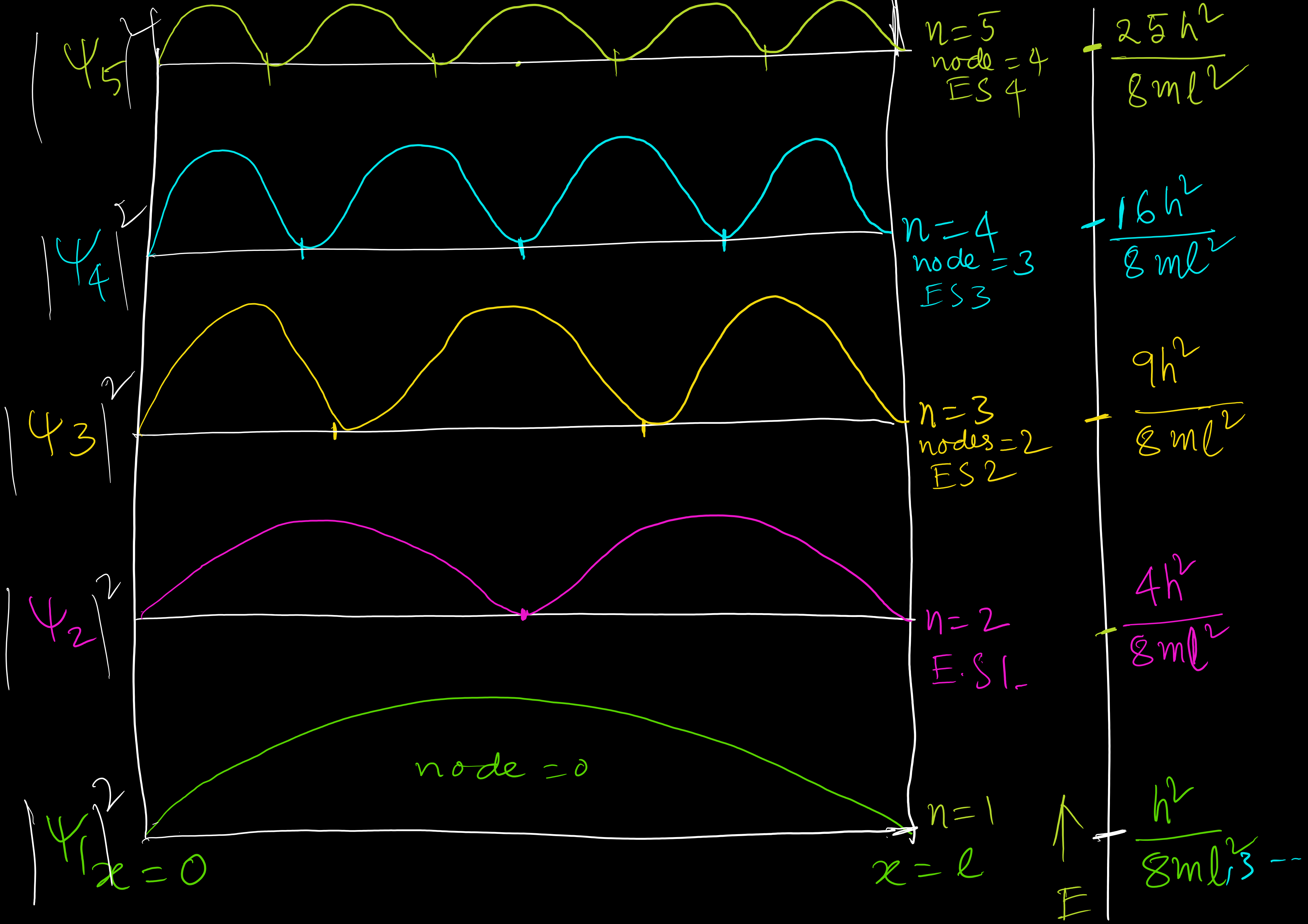
$$\text{Nodes} = (n-1) = 10-1 = 9$$

- First curve will be increasing



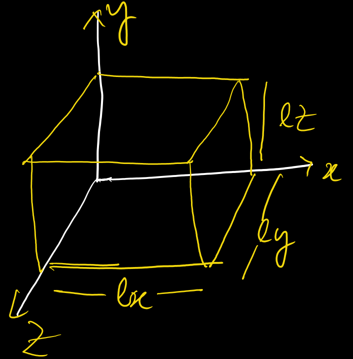
Probability of $\psi(x)$

$$E_n = \frac{h^2 n^2}{8ml^2}$$



Particle in 3D Box

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)=E\psi(x,y,z) \quad \text{— (1)}$$



$$\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi(x,y,z)=E\psi(x,y,z)$$

3D Schrodinger Equ^n

$0\le i\le l_i \qquad i=x,y,z$

$\psi(x,y,z)$ satisfies boundary condition $\Rightarrow$ vanishes at wall

$\psi(0,y,z)=\psi(l_x,y,z)=0$ for all y and z

$\psi(x,0,z)=\psi(x,l_y,z)=0$ for all x and z

$\psi(x,y,0)=\psi(x,y,l_z)=0$ for all x and y

We will use method of separation of variables to solve equ^n (1)

$$\psi(x,y,z)=X(x)\,Y(y)\,Z(z) \quad \text{— (2)}$$

Substituting eq^n (2) in (1) and then divide through by $\psi(x,y,z)=X(x)Y(y)Z(z)$ $\Rightarrow$

$$-\frac{\hbar^2}{2m}\frac{1}{X(x)}\frac{d^2X}{dx^2}-\frac{\hbar^2}{2m}\frac{1}{Y(y)}\frac{d^2Y}{dy^2}-\frac{\hbar^2}{2m}\frac{1}{Z(z)}\frac{d^2Z}{dz^2}=E$$

$$E_x+E_y+E_z=E$$

$$-\frac{\hbar^2}{2m}\frac{d^2X}{dx^2}=E_xX(x)\Rightarrow X_{n_x}(x)=A_x\sin\left(\frac{n_x\pi x}{l_x}\right)$$

$$-\frac{\hbar^2}{2m}\frac{d^2Y}{dy^2}=E_yY(y)\Rightarrow Y_{n_y}(y)=A_y\sin\left(\frac{n_y\pi y}{l_y}\right)$$

$$-\frac{\hbar^2}{2m}\frac{d^2Z}{dz^2}=E_zZ(z)\Rightarrow Z_{n_y}(z)=A_z\sin\left(\frac{n_z\pi z}{l_z}\right)$$

$$\int_0^{l_x}dx\int_0^{l_y}dy\int_0^{l_z}dz\,\psi^*(x,y,z)\psi(x,y,z)=1$$

$$A=A_xA_yA_z=\left(\sqrt{\frac{2}{l_x}}\right)\left(\sqrt{\frac{2}{l_y}}\right)\left(\sqrt{\frac{2}{l_z}}\right)=\sqrt{\frac{8}{l_xl_yl_z}}$$

$$\boxed{\psi_{n_xn_yn_z}(x,y,z)=\sqrt{\frac{8}{l_xl_yl_z}}\sin\left(\frac{n_x\pi x}{l_x}\right)\sin\left(\frac{n_y\pi y}{l_z}\right)\text{Sin}(n_z\text{Pi}\,z/l_z)}$$

$$\boxed{E_{n_xn_yn_z}=\frac{h^2}{8m}\left[\frac{n_x^2}{l_x^2}+\frac{n_y^2}{l_y^2}+\frac{n_z^2}{l_z^2}\right]}$$

$$\left.\begin{matrix}n_x\\n_y\\n_z\end{matrix}\right\}1,2,3\ ----$$

Zero point energy $\rightarrow$ Energy at G.S. $n=1$

$E \Rightarrow$ Additive

$\psi \Rightarrow$ Multiplicative

1D Box, $E_x = \dfrac{n^2h^2}{8ml_x^2}$, Zero point energy $(n=1)$ $= \dfrac{1^2h^2}{8ml^2} = \dfrac{h^2}{8ml^2}$

3D Box $E_T = E_x + E_y + E_z$

$$= \frac{n_x^2h^2}{8ml_x^2} + \frac{n_y^2h^2}{8ml_y^2} + \frac{n_z^2h^2}{8ml_z^2} = \frac{n_x^2h^2}{8ml^2} + \frac{n_y^2h^2}{8ml^2} + \frac{n_z^2h^2}{8ml^2}$$

For cubic box $l_x = l_y = l_z = l$

$$E = \frac{h^2}{8ml^2}(n_x^2 + n_y^2 + n_z^2)$$

$n_x \to 1$, $n_y \to 1$, $n_z \to 1$

$$\boxed{\text{So, zero point energy (ZPE)} = 3 \times \frac{h^2}{8ml^2}}$$

$$\boxed{E_{ZPE}(3D) = 3\,\underbrace{E_{ZPE}(1D)}_{h^2/8ml^2}}$$

Degeneracy

States having equal energy

If $l_x = l_y = l_z = l \rightarrow$ Cube

$$E_{n_x n_y n_z} = \frac{h^2}{8ml^2}\left(n_x^2 + n_y^2 + n_z^2\right)$$

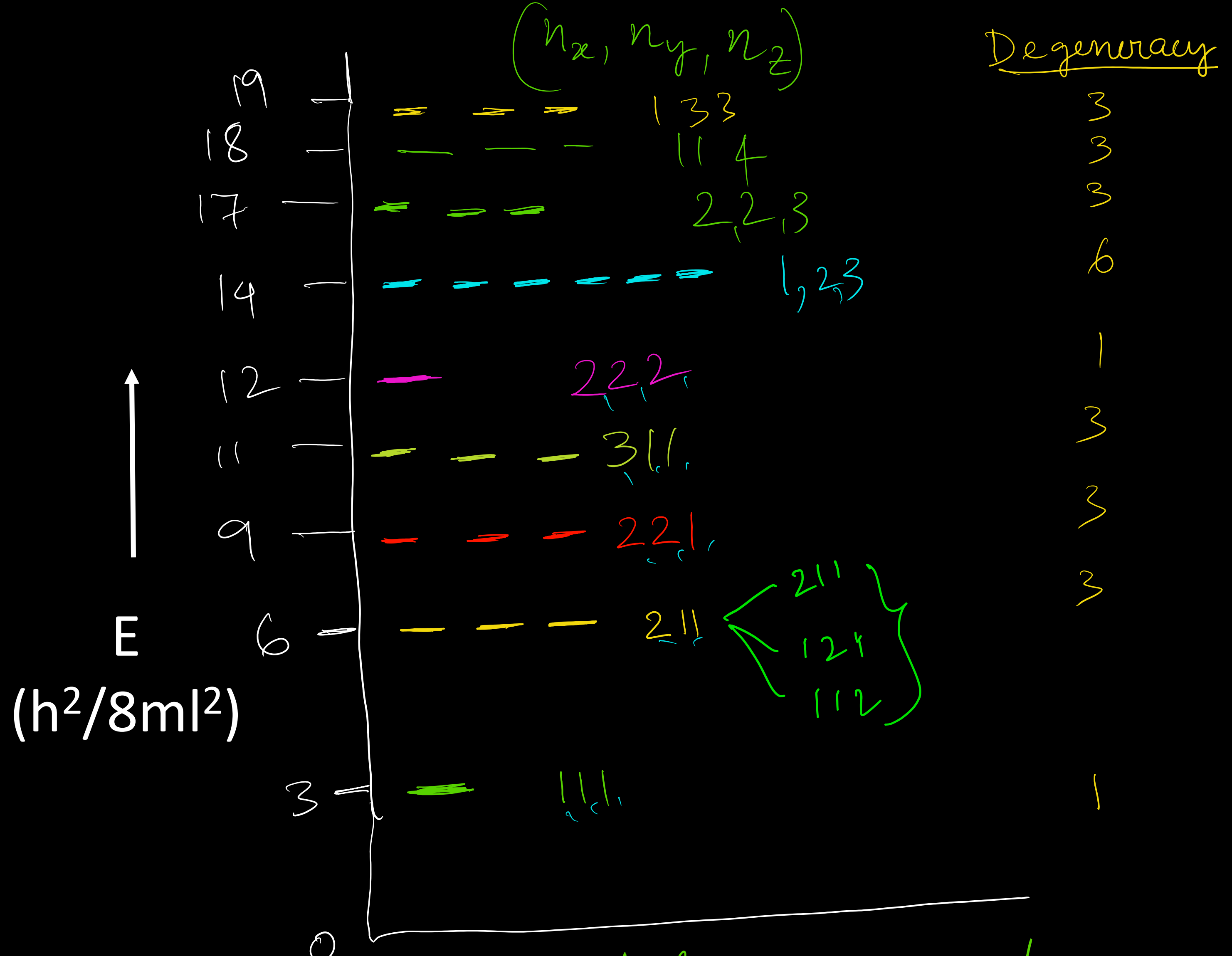
Symmetry $\rightarrow$ Degeneracy

E_{111}

$E_{112} \,/\, E_{211} \,/\, E_{121}$

$\underline{1\ 2\ 3}$

$$\left\{\begin{matrix} 1\ 2\ 3 \\ 1\ 3\ 2 \\ 2\ 1\ 3 \\ 2\ 3\ 1 \\ 3\ 1\ 2 \\ 3\ 2\ 1 \end{matrix}\right.$$



Energy levels for a particle in a cube.

Problems

- Prove that average momentum of a particle in a box is zero.

$$\langle p \rangle = 0$$